

Muhammad Ammar

AI / ML Engineer

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PROFILE

AI / ML engineer passionate about building intelligent systems and solving real-world problems using machine learning and data-driven approaches.

Experienced in developing end-to-end machine learning pipelines including data preprocessing, model training, model evaluation, and deployment.

Strong interest in computer vision and applied AI systems with hands-on experience building real-time and production-oriented ML solutions.

EDUCATION

Mansoura University | Faculty of Computer and Information Science
Bachelor's Degree in Computer Science

2022 – 2026

EXPERIENCE

• **Machine Learning Intern** , Cellula Technology

Oct 2025 – Dec 2025

SKILLS

Technical Skills:

- **Programming Skills:** Python, SQL
- **ML & AI:** Supervised Learning, Unsupervised Learning, Deep Learning, Computer Vision, CNNs, Transfer Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **Frameworks:** PyTorch, FastAPI, OpenCV
- **Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
- **Deployment:** Docker, ONNX, REST APIs
- **Version Control Systems:** Git, GitHub
- **OS:** Linux, Windows
- **Other:** Data Cleaning, Exploratory Data Analysis (EDA)

Soft Skills:

- **Languages:** English, Arabic
- **Other Skills:** Communication, analytical thinking, team collaboration, time management, problem solving

PROJECTS

- **University Graduation Project:** [Real-Time Industrial Defect Detection System](#)

Technologies: YOLO, OpenCV, FastAPI, WebRTC, Firebase, ONNX, TensorRT

- Developed a real-time computer vision system for automated defect detection in production lines.
- Implemented a two-stage pipeline combining box detection and tracking followed by defect detection.
- Optimized inference using ONNX and TensorRT for high-performance real-time processing.
- Built a dashboard using HTML, CSS and JavaScript with WebRTC support for live annotated camera streams.

- [Sign Language Recognition System](#)

Technologies: MediaPipe, OpenCV, PyTorch

- Developed a real-time sign language recognition system using webcam input.
- Implemented hand landmark detection and gesture classification pipeline.

- [AI Body Measurement System](#)

Technologies: Computer Vision, Segmentation, Python

- Built a computer vision system capable of estimating human body measurements from images.
- Used segmentation techniques and visual features to predict clothing measurements.

- [NYC Taxi Trip Duration Prediction](#)

Technologies: Python, Scikit-learn, FastAPI

- Built a regression model to predict taxi trip duration based on location and temporal features.
- Implemented feature engineering including haversine distance and time features.
- Served the trained model through a FastAPI prediction API.

- [Road Accident Risk Prediction](#)

Technologies: Python, Scikit-learn

- Developed a regression model using HistGradientBoostingRegressor to estimate accident risk.
- Implemented preprocessing pipelines and feature scaling for tabular data.
- Built training and prediction scripts for batch risk estimation.

- [Bank Marketing Campaign Prediction](#)

Technologies: Python, Scikit-learn, FastAPI

- Developed a machine learning pipeline to predict client subscription to term deposits using a Random Forest classifier.
- Implemented data preprocessing and handled class imbalance using SMOTE.
- Built a FastAPI service for real-time prediction through REST API.

COURSES & CERTIFICATIONS

- **Machine Learning Specialization** – deeplearning.ai
- **Deep Learning Specialization** – deeplearning.ai
- **Introduction to Machine Learning in Production** – deeplearning.ai
- **Machine Learning Diploma** – CSKilled